

HITIK KUMAR NAYAK

Junior Research Fellow (JRF) — VLSI & ASIC Design — Electronics & Communication Engineer

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🌐 [LinkedIn](#)

🐙 [GitHub](#)

🌐 [Portfolio](#)

TECHNICAL SKILLS

EDA & VLSI Tools: Cadence Virtuoso Suite (Schematic Capture, Layout XL), Mentor Graphics Calibre (nmDRC, LVS, PEX), LTspice, OriginPro, XCCircuit.

Analog IC Design: Two-Stage CMOS Op-Amp Architecture, Transistor Sizing (W/L), Differential Amplifiers, Bias Circuitry, Miller Frequency Compensation, Loop Stability Analysis, Noise, Common-Mode Rejection Ratio (CMRR) Optimization.

Physical Verification & Tape-Out: Symmetrical Layout Placement, Device Matching (Common-Centroid, Cross-Coupling), Guard Rings, Dummy Layout Insertion, Parasitic-Aware Layout Routing, I/O Pad Ring Integration, Automated Fill Generation, SCL 180 nm ASIC Flow.

Programming & Frameworks: Python, MATLAB, HTML, CSS, JavaScript, LaTeX (Advanced Technical Writing), Git/GitHub Version Control.

PROFESSIONAL EXPERIENCE

Junior Research Fellow (JRF)

ABV-IIITM Gwalior

Dec 2025 – July 2026

Gwalior, India

MeitY Sponsored C2S Project – Implantable Pacemaker Chip (iPACE-CHIP)

- Developed ultra-low-power analog/mixed-signal circuit blocks for an implantable pacemaker chip using Cadence, performing schematic design, custom layout, DRC/LVS verification, and physical design validation for MPW tape-out.

Graduate Engineer Trainee (Electrical)

MSP Steel & Power Ltd.

Jun 2025 – Jul 2025

Raigarh, India

- Monitored industrial substations, power distribution systems, and automated electrical controls while performing system diagnostics and preparing preventive maintenance documentation for high-voltage electrical infrastructure.

PROJECTS

Custom Two-Stage CMOS Op-Amp ASIC Design & Tape-Out

🐙 [GitHub](#)

- Designed a differential op-amp with schematic testbenches for CMRR/ICMR, developed custom common-centroid layouts with dummy structures, and achieved Calibre nmDRC/LVS sign-off and PEX post-layout simulations.
- Integrated the block into an I/O Pad Ring with seal rings for a complete GDSII tape-out flow using the SCL 180 nm process.

g_m/I_D -Based MOSFET Characterization and Transistor Sizing

🐙 [GitHub](#)

- Developed a g_m/I_D lookup table for MOS transistor characterization using Cadence Virtuoso and ADE L in a 180 nm CMOS process.
- Performed DC and parametric sweeps of V_{GS} , V_{DS} , and transistor width W to characterize MOS device behavior.
- Generated and analyzed g_m/I_D versus V_{OV} and g_m/I_D versus I_D/W characteristics for operating-point selection and transistor sizing.
- Used lookup-table data to identify suitable MOS bias points and required transistor dimensions for analog CMOS circuit design.

Summer Research Internship: 20T Hybrid Full Adder

🐙 [GitHub](#)

- Designed and simulated a low-power 20T Hybrid Full Adder in LTspice using conventional CMOS and FinFET technologies, analyzing short-channel effects, leakage current, and Power-Delay Product (PDP) optimization.

Major Project: Channel Capacity Optimization of 6G System

🐙 [GitHub](#)

- Scripted analytical channel simulators in MATLAB to evaluate communication behavior across sub-THz allocations (50–200 GHz).
- Modeled localized atmospheric constraints and molecular absorption variables to maximize spectral efficiency metrics against varying system operational distances.

INTERNSHIP EXPERIENCE

Intern – Electronics & Communication Engineering Department
National Institute of Technology (NIT) Raipur

May 2024 – Jun 2024
Raipur, India

Supervisor: Dr. Chitrakant Sahu (Associate Professor)

- Modeled and validated performance profiles for Hybrid Full Adders and standard sub-micron CMOS inverter topologies using LTspice.
- Simulated multi-gate 3D FinFET structures to study sub-threshold logic swing, absolute Power-Delay Product (PDP), and Fan-out-of-4 (FO-4) performance degradation boundaries across physical scale translations.

EDUCATION

Master of Technology – Microelectronics & VLSI
Indian Institute of Technology Bhilai, Durg, Chhattisgarh

2026-2028

Bachelor of Technology – Electronics & Communication Engineering
Guru Ghasidas Vishwavidyalaya, Bilaspur, Chhattisgarh

2021 – 2025
CGPA: 7.62 / 10.0

CERTIFICATIONS & ACHIEVEMENTS

- **GATE 2026 Qualified:** Electronics & Communication Engineering (ECE) — Score: 434.
- **TCS NQT Cognitive Qualification:** Dec 2025 — National Percentile Score: 73.66%.